```sql
CREATE TABLE STUDENT (
 Student_ID INT,
 Student_Name VARCHAR(50),
 Class VARCHAR(100),
 Marks INT,
 Fee_Paid DECIMAL(10,2),
 Date_of_Birth DATE,
 Admission_Date DATE
);

INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth,
Admission_Date) VALUES
(101, 'priyanshu', '10A', 85, 50000.00, '2007-06-05',
'2023-06-10'), (102, 'satvi', '10A', 92, 50000.00, '2008-11-20',
'2023-06-12'), (103, 'modi', '10B', 76, 45000.00, '2009-02-10',
'2023-06-11'), (104, 'amit', '10B', 88, 48000.00, '2008-08-25',
'2023-06-15'), (105, 'priya', '10A', 65, 40000.00, '2009-06-05',
'2023-06-13');
```

4. **Implement SQL Functions: Conversion Functions, and Date Functions.**

```sql
SELECT CAST('2025' AS UNSIGNED) + 5 AS Cast_Result;

SELECT CONVERT('123', SIGNED) + 7 AS Convert_Result;

SELECT Student_Name, CAST(Fee_Paid AS SIGNED) AS Fee_Integer
FROM STUDENT;

SELECT Student_Name, CAST(Marks AS CHAR) AS Marks_Text
FROM STUDENT;

SELECT STR_TO_DATE('15-08-2024', '%d-%m-%Y') AS Converted_Date;

SELECT Student_Name, DATE_FORMAT(Admission_Date, '%d %b %Y') AS
Admission_Formatted
FROM STUDENT;
SELECT Student_Name, FORMAT(Fee_Paid, 2) AS Fee_Formatted
FROM STUDENT;

-- Date functions
SELECT Student_Name,
       YEAR(Date_of_Birth) AS Birth_Year,
       MONTH(Date_of_Birth) AS Birth_Month,
       DAY(Date_of_Birth) AS Birth_Day,
```

```
        DAYNAME(Date_of_Birth) AS Birth_Weekday
FROM STUDENT;

SELECT Student_Name,
        Admission_Date,
        DATE_ADD(Admission_Date, INTERVAL 1 YEAR) AS One_Year_Later,
DATE_SUB(Admission_Date, INTERVAL 15 DAY) AS Fifteen_Days_Before FROM
STUDENT;

SELECT Student_Name,
        DATEDIFF(CURDATE(), Admission_Date) AS Days_Since_Admission,
        TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT CURDATE() AS Today, LAST_DAY('2024-02-10') AS Month_End;
```

Output:

Student_Name	Admission_Date	One_Year_Later	Fifteen_Days_Before
Priyanshu	2023-06-10	2024-06-10	2023-05-26
Satvi	2023-06-12	2024-06-12	2023-05-28
Modi	2023-06-11	2024-06-11	2023-05-27
Amit	2023-06-15	2024-06-15	2023-05-31
Priya	2023-06-13	2024-06-13	2023-05-29

Student_Name	Days_Since_Admission	Age
Priyanshu	1210	19
Satvi	1208	17
Modi	1209	17
Amit	1205	18
Priya	1207	17

Today	Month_End
2026-10-02	2024-02-29

5. **Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators.**

```
SELECT 15 + 4 AS Addition, 15 - 4 AS Subtraction, 15 * 4 AS
Multiplication,
       15 / 4 AS Division, 15 DIV 4 AS Int_Division, 15 % 4 AS Modulus;
```

```
SELECT Student_Name, Marks,
       Marks + 5 AS Marks_Plus_5,
       Marks * 2 AS Marks_Double,
       Fee_Paid / 12 AS Monthly_Fee,
       Marks % 10 AS Marks_Mod_10
FROM STUDENT;

-- Comparison
SELECT Student_Name, Marks FROM STUDENT WHERE Marks > 80;
SELECT Student_Name, Marks FROM STUDENT WHERE Marks < 76;
SELECT  Student_Name,  Marks  FROM  STUDENT  WHERE  Marks  >=
88;
SELECT Student_Name, Marks FROM STUDENT WHERE Marks <= 76;
SELECT Student_Name, Class FROM STUDENT WHERE Class = '10A';
SELECT Student_Name, Class FROM STUDENT WHERE Class <>
'10A';

SELECT 5 = 5 AS Equal, 5 <> 3 AS Not_Equal, 5 > 8 AS Greater, 5 <=> 5
AS Null_Safe_1, NULL = NULL AS Null_Eq, NULL <=> NULL AS Null_Safe_2;

-- Logical
SELECT Student_Name, Class, Marks
FROM STUDENT
WHERE Class = '10A' AND Marks > 80;

SELECT Student_Name, Class, Marks
FROM STUDENT
WHERE Class = '10B' OR Marks > 90;

SELECT Student_Name, Marks
FROM STUDENT
WHERE NOT (Marks >= 76);

SELECT Student_Name, Class, Marks
FROM STUDENT
WHERE (Class = '10A') XOR (Marks > 80);
```

6. **Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples Special Operators, and Set Operations(UNION)**

```
CREATE TABLE Sports_Club (Student_Name VARCHAR(50), Class VARCHAR(100));
CREATE TABLE Music_Club (Student_Name VARCHAR(50), Class VARCHAR(100));
```

```sql
INSERT INTO Sports_Club VALUES ('priyanshu','10A'), ('satvi','10B'),
('priya','10A');
INSERT INTO Music_Club VALUES ('priya','10A'), ('amit','10B'),
('modi','10B');

-- Special operators
SELECT Student_Name, Marks FROM STUDENT WHERE Marks BETWEEN 70 AND 90;

SELECT Student_Name, Class FROM STUDENT WHERE Class IN ('10A');

SELECT Student_Name, Marks FROM STUDENT WHERE Marks NOT IN (65, 92);

SELECT Student_Name FROM STUDENT WHERE Student_Name LIKE 'S%';

SELECT Student_Name FROM STUDENT WHERE Student_Name LIKE '_h%';

SELECT Student_Name FROM STUDENT WHERE Student_Name LIKE '%th';

SELECT Student_Name, Fee_Paid FROM STUDENT WHERE Fee_Paid IS NOT

NULL;

SELECT Student_Name, Marks
FROM STUDENT
WHERE Marks > ALL (SELECT Marks FROM STUDENT WHERE Class = '10B');

SELECT Student_Name, Marks
FROM STUDENT
WHERE Marks > ANY (SELECT Marks FROM STUDENT WHERE Class = '10B');

-- Set operations
SELECT Student_Name, Class FROM Sports_Club
UNION
SELECT Student_Name, Class FROM Music_Club;

SELECT Student_Name, Class FROM Sports_Club
UNION ALL
SELECT Student_Name, Class FROM Music_Club;
```

Converted_Date
2024-08-15

Student_Name	Admission_Formatted
Priyanshu	10 Jun 2023
Satvi	12 Jun 2023
Modi	11 Jun 2023
Amit	15 Jun 2023
Priya	13 Jun 2023

Student_Name	Fee_Formatted
Priyanshu	50,000.00
Satvi	50,000.00
Modi	45,000.00
Amit	48,000.00
Priya	40,000.00

Student_Name	Birth_Year	Birth_Month	Birth_Day	Birth_Weekday
Priyanshu	2007	6	5	Tuesday
Satvi	2008	11	20	Thursday
Modi	2009	2	10	Tuesday
Amit	2008	8	25	Monday
Priya	2009	6	5	Friday

Student_Name	Admission_Date	One_Year_Later	Fifteen_Days_Before
Priyanshu	2023-06-10	2024-06-10	2023-05-26
Satvi	2023-06-12	2024-06-12	2023-05-28
Modi	2023-06-11	2024-06-11	2023-05-27
Amit	2023-06-15	2024-06-15	2023-05-31
Priya	2023-06-13	2024-06-13	2023-05-29

Student_Name	Days_Since_Admission	Age
Priyanshu	1210	19
Satvi	1208	17
Modi	1209	17
Amit	1205	18
Priya	1207	17

Today	Month_End
2026-10-02	2024-02-29

Addition	Subtraction	Multiplication	Division	Int_Division	Modulus
19	11	60	3.7500	3	3

Student_Name	Marks	Marks_Plus_5	Marks_Double	Monthly_Fee	Marks_Mod_10
Priyanshu	85	90	170	4166.666667	5
Satvi	92	97	184	4166.666667	2
Modi	76	81	152	3750.000000	6
Amit	88	93	176	4000.000000	8
Priya	65	70	130	3333.333333	5

Student_Name	Marks
Priyanshu	85
Satvi	92
Modi	76
Amit	88
Priya	65

https://github.com/priyanshusahoo021/-data-base-management-system.git